

Harpreet Singh

FPGA / HLS / Hardware Acceleration Engineer

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Summary

FPGA and embedded AI engineer focused on Vitis HLS, Vivado, MicroBlaze, Zynq, AXI4-Lite, AXI HP, BRAM data paths, fixed-point neural accelerators, and UART-visible board validation. Built research pipelines for QSNN, QiSNN, SNN, QLIF, PC-DDM-SNN, ANN baselines, N-MNIST, MNIST14, rodent-raster decision decoding, and Zynq-based LLM acceleration.

Core Skills

- Xilinx Vivado, Vitis HLS, HLS C/C++, MicroBlaze, Zynq Cortex-A9, AXI4-Lite, AXI HP, BRAM maps, UARTLite
- Arty A7-35T, Arty A7-100T, Arty Z7-20, Artix-7, Zynq-7000, XSA, bitstream, Vitis platform, board bring-up
- Fixed-point arithmetic, `ap_fixed<16,6>`, `ap_fixed<12,4>`, quantized weights, INT8/INT4 matrix-vector acceleration
- ANN, SNN, QSNN, QiSNN, QLIF, PC-DDM-SNN, temporal N-MNIST, MNIST14, rodent raster, early-exit inference
- Python, NumPy, PyTorch, TensorFlow/Keras, model export, HLS testbenches, UART test automation, robustness evaluation

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | University of Lethbridge | Apr 2025 - Present

- Build FPGA research pipelines for ANN, SNN, QSNN/QiSNN, QLIF, PC-DDM-SNN, and rodent-raster decision experiments using Vitis HLS, Vivado, MicroBlaze, BRAM windows, and UART diagnostics.
- Developed dataset-to-BRAM flows using 196-feature MNIST14 inputs and 2352-word temporal N-MNIST/rodent-raster inputs with fixed-point payloads and software reference checks.
- Documented HLS/Vivado/Vitis rebuild flows, XSA/bitstream paths, UART tests, timing/resource evidence, and hardware-claim boundaries.

Research Assistant | University of Regina | Jan 2021 - Apr 2024

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECUREPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | Self-Employed | Jun 2024 - Mar 2025

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | Planetcast Media Services Ltd. | Dec 2018 - Nov 2020

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

Embedded Software Engineer / R&D Engineer | Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd. | Feb 2016 - Dec 2018

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

Selected Projects

QSNN_EarlyExit_FPGA_Research

- Built an Arty A7-35T QSNN MicroBlaze system with AXI4-Lite control, BRAM input/output buffers, UART diagnostics, and MNIST board tests; recorded 0.698-0.699 ms HLS latency at 100 MHz, 9.87% LUTs, 5.13% FFs, 41.0% BRAM tiles, 3.33% DSPs, zero routing errors, and positive routed timing slack.

QLIF_QSNN_FPGA

- Created PC-DDM-SNN and Quantum LIF software-to-hardware flows for MNIST/N-MNIST, including dataset-matched training, HLS weight export, Vitis HLS, Vivado, XSA/bitstream production, and 1000-sample UART test workflows.

snn_qisnn_rodent_raster

- Adapted temporal QiSNN/SNN FPGA pipelines for rodent neural raster windows with 2352-word input BRAM, AXI4-Lite `ap_ctrl_hs`, fixed-point payloads, MicroBlaze readback, and debug-visible state planning.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRIPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012